SEASUG Paper #67-2026

The Analysts Guide to Efficient Theme Park Tourism

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ABSTRACT

Theme parks have been the pinnacle of excitement for many tourists for decades. The expectation of a theme park is not only about seeking thrill and enjoying multiple attractions, but from the overall experience of the environment as a whole. Tourists look forward to visiting as many themed attractions as possible, but this could present an issue when accounting for factors such as distance and wait times between each ride. Seeking the optimal route to enjoy each attraction possible while reducing wait times is something each and every attendee wants. However, what is the most optimal way to achieve this goal? In this paper, we analyze the wait time distributions of rides at Disneyland in Anaheim, California, and extrapolate categories of attractions. This can provide parkgoers with helpful information on what to expect when going to Disneyland, and how to plan to best use their time.

INTRODUCTION

With over 1,200 amusement and theme parks across the United States as of the year 2023, attraction-based parks are an American staple [1]. As the industry has grown, and ticket prices for “single destination” theme parks, like Disneyland, grow with it, it has never been more important for a parkgoer to optimize their time at a theme park. With more to do and varying ticket prices including premium options like “fast-lanes,” getting the most out of a trip to a park is a difficult task an entire industry has grown out of. From blogs to forums to purchasable software, there is a plethora of information to digest for planning a weekend trip to a park like Disneyland or Universal Studios. There is value in taking the time to put together a thoughtful itinerary, but for those that do not have time to dedicate to detailed planning may want general “rules of thumb” to help make their trip better. By conducting exploratory data analysis, we can analyze trends in wait time data to create guidelines for how to best utilize time inside a park and avoid waiting in line unnecessarily.

METHODS AND VISUALIZATION

The focus of this work is in creating a simple guide for visiting Disneyland, without requiring a park attendee to create a detailed itinerary, allowing for a single rider or group to make informed routing decisions on their own. The main way for determining these best practices is through visual analysis in exploratory data analysis. The data analyzed here was obtained from *queue-times.com*, a site that collects 5-minute interval wait times for theme parks across the world [3]. One year of data was collected from January 1st to December 31st, 2023, by webscraping historic wait times. All rides operated for at least 85% of time periods across the year (n = 31). By visualizing different facets of a park and its rides, we can see how wait times change over time and extrapolate when rides are likely to be at a local minimum wait time. Looking at seasonality gives even better insight, so a parkgoer can adjust their planning based on the time of year they go to the park. We add to this by analyzing how average ride wait times change per ride over the span of a year. This allows us to categorize rides based on similar wait time distributions and provide guidelines for when to plan a trip

season	avg_wait_time
Net Avg	21.642855246
Fall	22.239349122
Spring	21.310886614
Summer	21.059706282
Winter	21.969845465

Table 1. Average Wait Time per Season

SEASONAL PLANNING

When deciding to visit a theme park like Disneyland, one of the first decisions that has to be made is: "When?" Table 1 indicates that the season with the highest wait time average is Fall, and that the Summer is the best season to plan a trip for, as it has the lowest average wait times. The differences in average wait between these seasons are small though and more can be gleaned from Table 2. The lowest average wait time for a day in a season is in August on Sundays, with an average wait time for the day of 14.6 minutes. Looking at this table more deeply provides the first rule of thumb for attending Disneyland, on average, wait times start high on Mondays, and decrease over the week to Sunday. There are some mid-week bumps around Thursday, but overall, there is a trend of decreasing across the week in most seasons. August, September and October being the exceptions where there is a meaningful increase from Friday to Saturday. This aligns with the idea that if a family or group of friends were to go to a theme park like Disneyland, their schedule is likely constrained by an adult who is working Friday. The natural conclusion of this is that many people will travel to the park on Saturday or Sunday and begin a week-long vacation at Disneyland on Monday. So, if possible, getting to the park later in the week, and leaving Sunday night or Monday morning provides lower wait times on average.

Average Disneyland Wait Times by Month and Weekday
Grouped by Season

		Weekday						
		Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Season	Month	Average Wait Time (min)	Average Wait Time (min)	Average Wait Time (min)	Average Wait Time (min)	Average Wait Time (min)	Average Wait Time (min)	Average Wait Time (min)
Fall	October	23.2	23.6	23.0	21.2	22.4	22.3	26.7
	November	18.5	24.6	23.5	21.5	24.6	20.7	19.8
	September	21.2	24.4	22.3	20.5	22.1	18.6	22.1
Spring	May	18.0	21.8	22.9	22.9	19.2	21.4	19.7
	April	16.6	23.6	22.8	23.1	21.4	22.1	20.8
	March	20.0	25.8	21.0	20.2	22.5	21.2	21.2
Summer	July	17.6	24.2	21.4	20.9	22.3	19.7	18.2
	June	17.9	24.6	23.7	24.4	23.3	22.2	21.2
	August	14.6	23.4	22.6	21.0	20.8	18.2	19.4
Winter	January	21.4	23.0	21.5	23.5	21.4	23.4	21.4
	December	21.7	20.3	20.6	21.8	21.8	23.3	23.7
	February	22.2	22.9	20.9	22.5	19.4	21.8	22.2

Table 2. Average Wait Time per Day of Week by Month

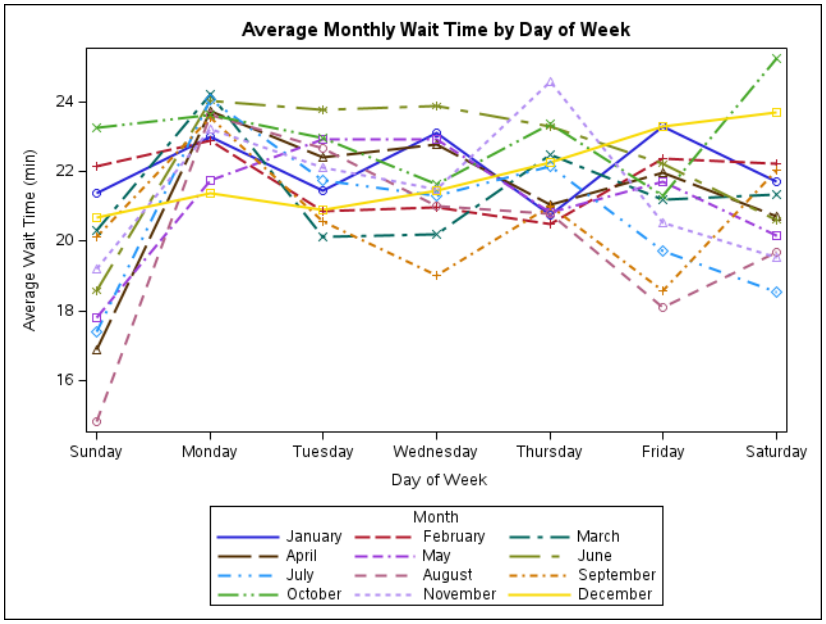


Figure 1. Average Monthly Wait Time by Day of Week

AVOIDING EVENTS AND HOLIDAYS

For selecting a better week to go to Disneyland, common knowledge would say we want to plan around weeks that have special events or holidays, as it is assumed those tend to drive traffic to the park. Whether events specific to the park, or nationwide events for the season. Actual wait time averages suggest that planning a trip leading up to, and on such holidays can result in a trip with decreased average wait times. Particularly for holidays associated with families and children. When looking at the plot of wait times in July over the week in Figure 2, we can see that most weeks in July follow the same trend of decreasing wait time over the week from Monday to Sunday, with slight variation in magnitude per week and direction from day to day. However, the week of July 2nd to July 8th has a spike around the 5th and 6th of July, Wednesday and Thursday, immediately following the national holiday of July 4th that parkgoers are likely to have other plans for.

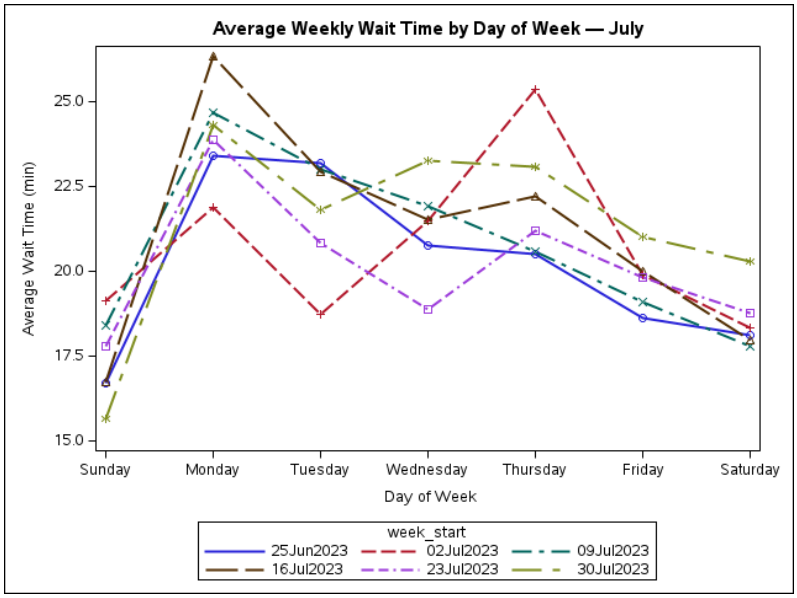


Figure 2. Average Weekly Wait Time by Day of Week – July

Another interesting example of this is the month of October in Figure 3. The month has a decreasing trend from Sunday to Friday, with some weeks having increasing wait times from Friday back to Sunday. The exception is the week of Halloween, where we see a sharp decline from Saturday the 28th to Tuesday the 31st, but there is a great increase for November 1st and 2nd. This lines up with the days following Halloween. Disney runs seasonal events for Halloween from August to October, but nothing special for Halloween itself. As a result, it is reasonable to think many families would have alternate plans for the holiday.

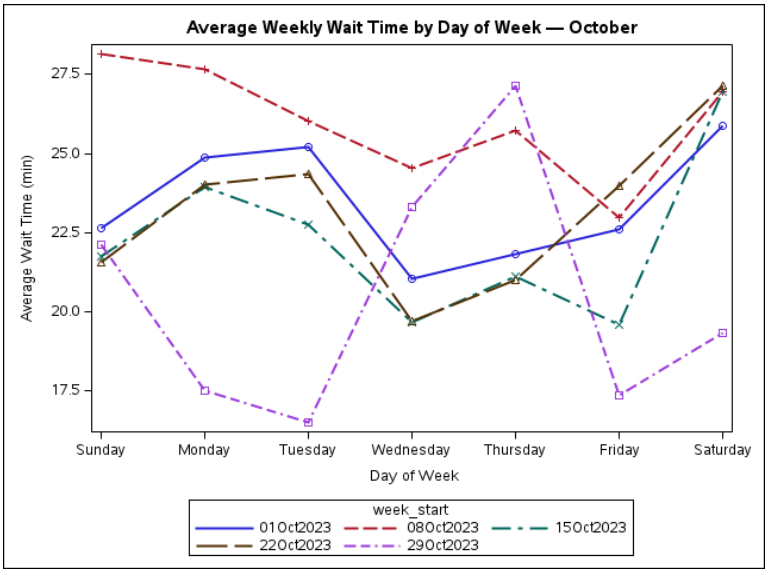


Figure 3. Average Weekly Wait Time by Day of Week – October

These post-holiday related spikes in wait time are consistent across multiple types of holidays, but in particular federal holidays are typically associated with time off. In February there is a high spike in wait time around President's Day from February 18th to 22nd. In May there are elevated wait times the weeks of Mother's Day (May 15th – 17th) of Memorial day (May 28th - June 03). In contrast to this, Labor Day weekend did not see much of an increase in average wait time around September 4th.

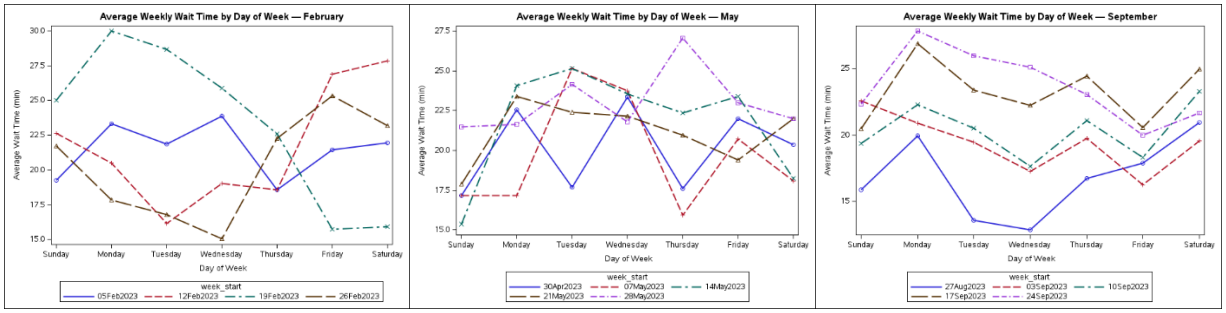


Figure 4. Average Weekly Wait Time by Day of Week – February, May, September

November is an interesting outlier from most months, the spike around Halloween (October 31st) leads directly into a Veterans' Day weekend the next week. The wait times increase during the week leading up to the holiday, but do not seem to raise the wait times much above the average for the weekend. This is noted as this could motivate planning around such elongated weekends for families looking to have a late fall weekend getaway.

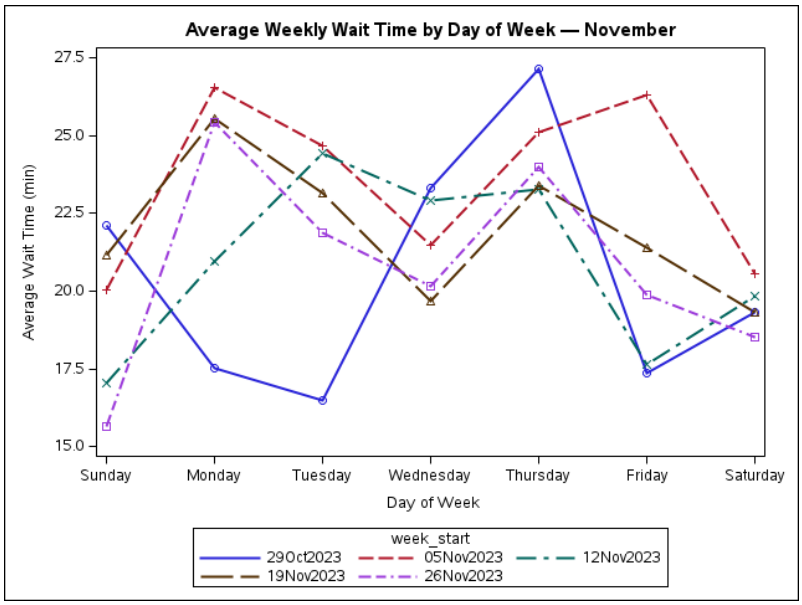


Figure 5. Average Weekly Wait Time by Day of Week – November

Across seasons, there is a difference in the consistency of wait time change over the week. In the summer season, weekly wait time has minor differences in magnitude and direction, with the plot of July 2023 being a great example of a month with little deviation from a near-linear decrease in wait time from Monday to Sunday. Starting around March, until August, wait times condense to this decreasing trend with slight variation. This pattern has less variation in the mid-summer months like June and July, and more variation around March and August. From October to March, wait times average out to mostly flat in direction, with wait times being higher from Friday to Monday. Months like January, February, and October, have most weeks following concave shaped wait averages with some outliers, such as around Halloween. November has a more consistent trend in wait time change, similar to the summer months, and December has almost constant wait times throughout the month.

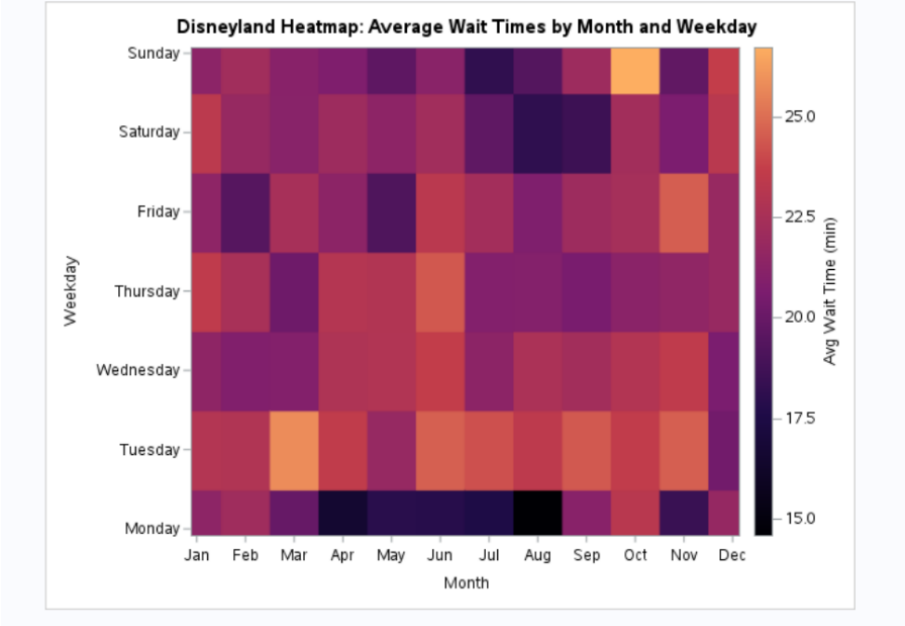


Figure 6. Weekly Disneyland Wait Time Heatmap

RIDE CATEGORIZATION

When looking at ride trends over the year, we find 4 general categories of ride wait time distribution: Low-Intensity Indoor Rides (i.e. *It's a Small World*, *Haunted Mansion Holiday*), Low-Intensity Outdoor Rides (i.e. *Dumbo the Flying Elephant*, *Jungle Cruise*), High-Intensity Experience-Focused Rides (i.e. *Space Mountain*, *Buzz Lightyear Astro Blasters*), Low-Intensity Experience-Focused Rides (i.e. *Snow White's Enchanted Wish*, *Alice in Wonderland*). Low-Intensity Indoor Rides are the smallest category, containing three rides marked by a trend of low wait times for a majority of the year with a sharp increase in wait times in the last four months of the year. These appear to be associated with seasonal themes, events, and colder temperatures. *It's a Small World*, an iconic and historic ride in the park, has an increase in average wait times by over 10 minutes from October to November. This is likely tied to Christmas theming that is added to the ride during these months through December. *Haunted Mansion Holiday* similarly shows wait time increase around September and October, as it too gets seasonal decor in these months. *Star Tours: The Adventure Continues* is the odd one out in this category, as there is an increase in the average wait time of about 7 minutes from July to August. This is likely linked to the August 2023 release of the television series *Ashoka* [2]. However, the other *Star Wars* themed rides saw a decrease in wait time during August. *Star Tours* is an outlier in this case but does indicate the possibility of show and movie releases meaningfully influencing wait times. Only *It's a Small World* has a queue that is majority outdoors, the other rides have queues that are indoors and likely heated/cooled, making waiting in queue more pleasant when temperatures are more extreme.

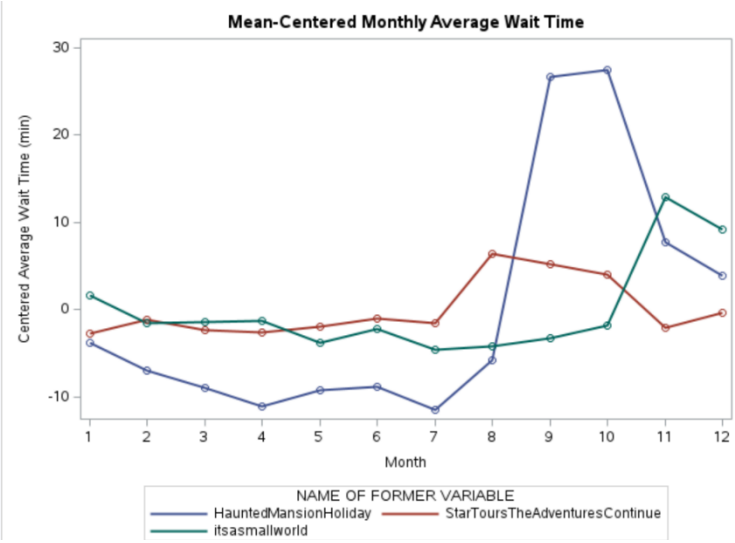


Figure 7. Mean-Centered Monthly Average Wait Time – Low-Intensity Indoor Category

Low-Intensity Outdoor Rides follow a trend of higher wait times in the spring and fall months, as well as specific peaks in June. Most peak around March, October, and June such as *King Arthur's Carrousel* and *Millennium Falcon: Smuggler's Run*. This is likely a result of these rides being more popular in seasons where temperatures are milder, while also being rides of high interest that see an increase in traffic during the busiest months around June. *Smuggler's Run* is a unique ride in this category as it is an entirely indoor ride, but it potentially follows this trend due to having much of the queue exposed to open air, thus making people less likely to queue when temperatures are at extremes. Also in this category are the three restaurants with wait-time information, *Cafe Orleans, Carnation Cafe*, and *Oga's Cantina*. *Oga's Cantina* in particular typifies this category with distinct peaks in April, July, and November. Four other rides, *Autopia*, *Astro Orbiter*, *Davey Crockett's Explorer Canoes*, and *Disneyland Monorail*, are in this category as well, while not perfectly fitting the trend. These have much stronger June/July spikes than other rides, with lower spikes in the spring and fall. *Autopia* in particular has no spikes in the spring or fall but does have a sharp wait time average increase in the month of December.

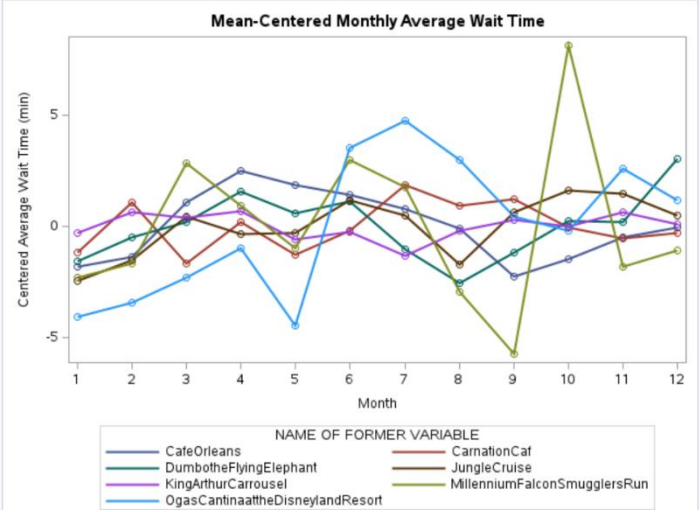


Figure 8. Mean-Centered Monthly Average Wait Time – Low-Intensity Outdoor Category

High-Intensity Experience-Focused Rides are ones highly tied to intellectual properties, like *Pirates of the Caribbean* or *Peter Pan's Flight*, as well as the rides with the highest traffic in the park such as *Space Mountain, Matterhorn Bobsleds,* and *Star Wars: Rise of the Resistance*. These rides follow the same trend as other categories with wait time spikes around June and October, but these tend to also have a winter season spike as well. In many cases, such as with *Star Wars: RotR* and *Buzz Lightyear: Astro Blasters*, these rides have almost uniform wait times throughout the year, with dips in wait time only in August.

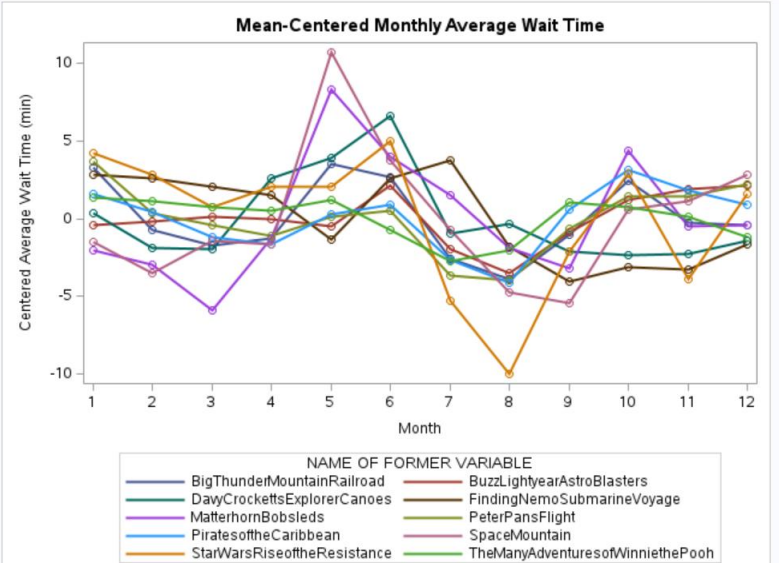


Figure 9. Mean-Centered Monthly Average Wait Time – High-Intensity Experience-Focused Category

Low-Intensity Experience-Focused Rides are those tied to intellectual properties like *Snow White's Enchanted Wish* and *Pinocchio's Daring Journey*. These tend to be "dark ride" attractions, typically single-track guided vehicle rides where parkgoers travel through scenes of fictional characters. The rides in this category which are not dark rides are aimed at children or are a similar low-intensity experience to dark rides, such as *Storybook Land Canal Boats*, which are a boat ride around the Fantasyland section of Disneyland. These rides have average wait times that are higher at the start of the year, and decrease into July and August, with some having small increases in wait times in June. Interestingly, all rides but *Disneyland Railroad* in this category are localized to the Fantasyland area of the park.

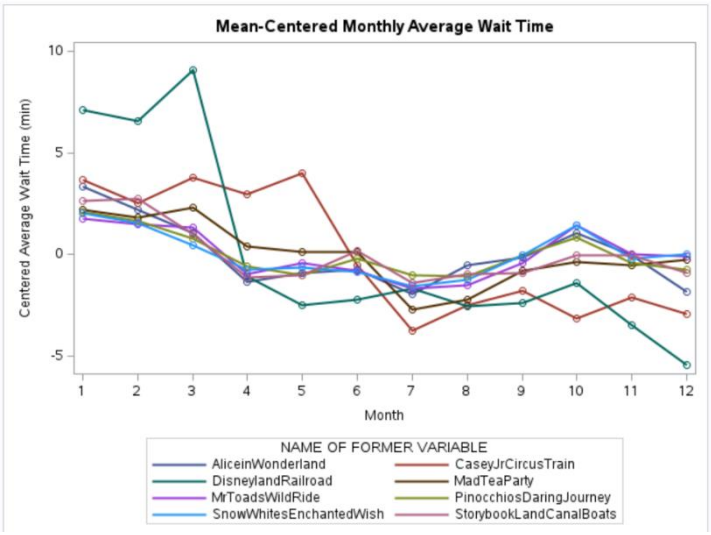


Figure 10. Mean-Centered Monthly Average Wait Time – Low-Intensity Experience-Focused Category

Knowing how these rides change over the year can help inform how parkgoers book trips to Disneyland based on rides they are likely to prioritize. Those that are fans of classic Disney intellectual properties are likely highly interested in riding the Low-Intensity Experience-Focused attractions and thus would have the best experience in regard to lowered wait times if they planned a trip in May, July or August, when a majority of rides in this category have some of their lowest average wait times. Those that are interested in the High-Intensity Experience-Focused rides as well as the Low-Intensity ones would find July and August to be a sweet spot for both, while May would be a worse option as High-Intensity rides have higher average wait times in May. Fans of Low-Intensity Outdoor rides will find January to be the best month to book a vacation if they can find a week with sunny days and higher temperatures. For warmer months, May has consistently lower wait times. This is in contrast to August being a lower wait time month on average. For this category in particular there is variance if the late-summer dip is in August, September, or October, and thus the best warm month for a consistently low wait time experience would be May. Finally, for Low-Intensity Indoor rides, any time of year prior to September avoids the consistent seasonal events for rides. They do additionally give some insight into a factor that is more difficult to predict: special events. Since these are highly tied to release schedules for different intellectual properties, being aware of what potential events are happening at a park is also crucial for avoiding booking a trip during a high-traffic week.

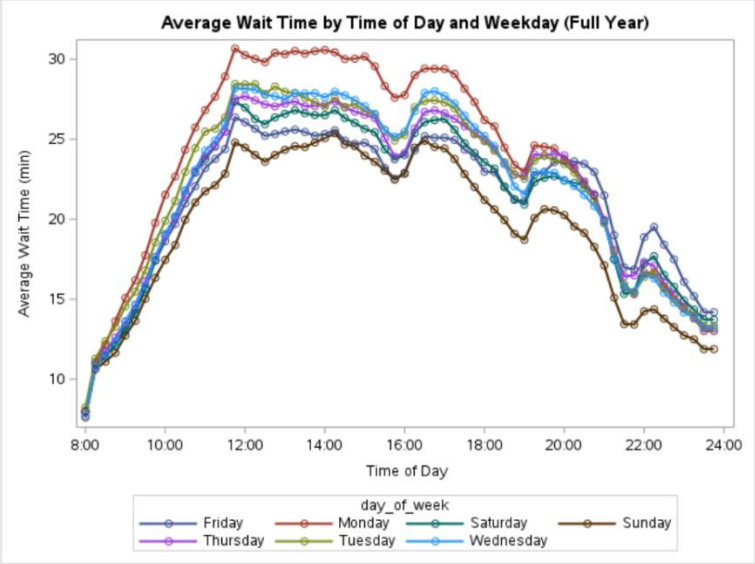


Figure 11. Yearly Average Wait Time by Time of Day and Weekday

HOW RIDE AVERAGE WAIT TIME CHANGES OVER A DAY

When looking at individual ride wait times over the average day, we can see dips in wait times that represent crowds entering and exiting the park. When planning a day at Disneyland, aiming to start waiting in lines when crowds start thinning is ideal. This occurs around 2pm to 4pm, 5pm to 7pm, and 8pm to close. These patterns suggest that an afternoon-to-late-night Disneyland trip is ideal for those wanting minimal wait times for rides. However, this time range may not be feasible for everyone, particularly families with young children. So, determining when it is right to ride specific rides during the day is highly useful.

The first set of rides we will look at are those with the highest average wait times in the park, *Star Wars: Rise of the Resistance*, *Space Mountain*, *Matterhorn Bobsleds*, *Big Thunder Mountain Railroad*, and *Peter Pan's Flight*. These are the top five rides in the park in terms of average yearly wait time, and when looking at the mean-centered average daily wait time distribution, we can see that these rides have very similar distributions that generally follow the park average distribution. With the exception of *Star Wars: RotR* and *Space Mountain*, these rides tend to have a constant wait time from 11am to 6pm, when wait times start to trail off and dip below their average. *Star Wars: RotR* breaks this trend by having a high early-morning peak, but trending downward as early as noon. *Space Mountain* does the opposite as it almost exclusively increases in average wait time from open to 5pm, where it generally holds a high average wait time until 8pm when it starts to trend downward. All rides in this plot experience a 9:30pm valley into a 10pm spike, indicating that there is a group of parkgoers that tend to arrive at night. For these top five wait time rides, the best time to ride them is at park open, or near park close, as once the park has filled up, they are likely to be busy and have high wait times for the majority of the day. But that also means that for those willing to wait in line, it is unlikely that the ride will have a seemingly random increase in wait time, as wait times tend to be generally constant throughout the day.

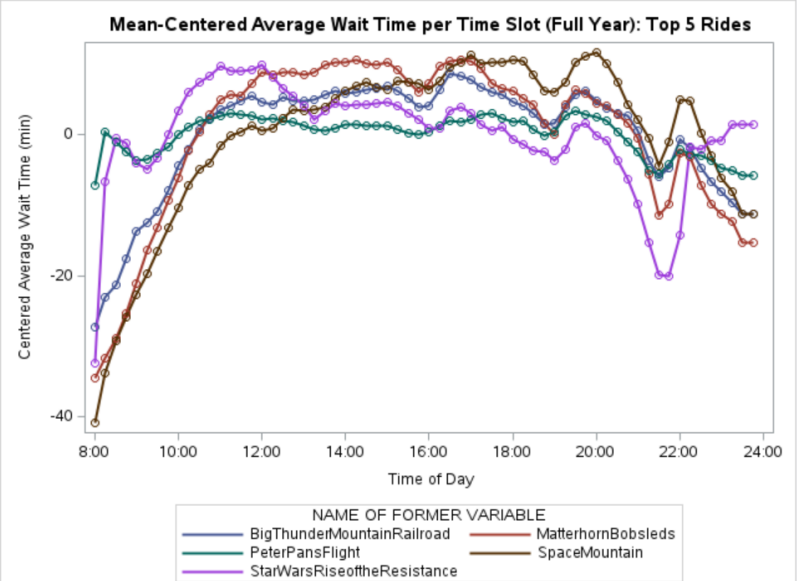


Figure 12. Mean-Centered Yearly Average Wait Time by Time of Day: Top 5 Rides

Disneyland is separated into multiple “lands” which have an overarching theme. This is of interest to this work as these rides being close in proximity and of a particular theme could potentially elucidate parkgoer behavior. The first of the two “lands” we look at is Fantasyland. A section of the park located in the center, immediately behind the *Sleeping Beauty* castle set piece. It is an area of the park that is highly likely to be visited first by an attendee as it is one of the first places that one sees when walking through Main Street, U.S.A., the entrance street to Disneyland. This can be seen in the mean-centered average plot as the major rides in this area have a near constant wait time throughout the day from 11AM to 8PM. Notably, *Peter Pan’s Flight*, is a ride that averages in the top 5 rides in the park for wait time and it follows the same trend as the rest of Fantasyland near identically. This indicates that, while *Peter Pan’s Flight* receives a high amount of traffic, many riders are likely to go to Fantasyland for this ride in particular. Although *Alice in Wonderland* is not far behind in average wait time.

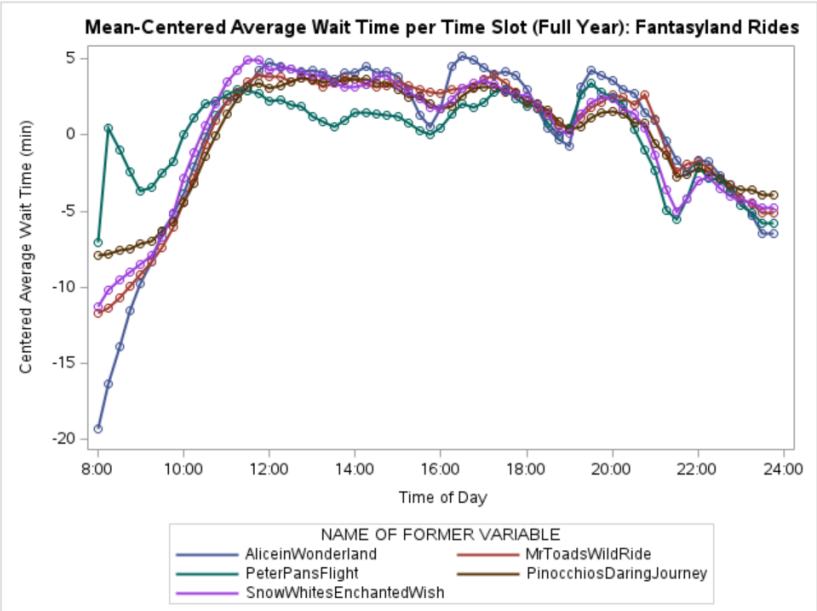


Figure 13. Mean-Centered Yearly Average Wait Time by Time of Day: Fantasyland Rides

In contrast to Fantasyland, Tomorrowland sees a more marked decrease in wait times as the day progresses. Tomorrowland is a section of the park with futuristic themes on the east side. While Fantasyland is eye-catching due to the *Sleeping Beauty* castle that fronts it, Tomorrowland is closer. Its first ride, *Astro Orbiter*, is practically inside of the Main Street, U.S.A. plaza, and is a major eye-catching set piece itself. The one ride that breaks the decreasing trend of this area is *Space Mountain*, one of the top five rides in terms of average wait time in the park, which actually increases in wait time as the day goes on. Only beginning to decrease in wait time around 8PM.

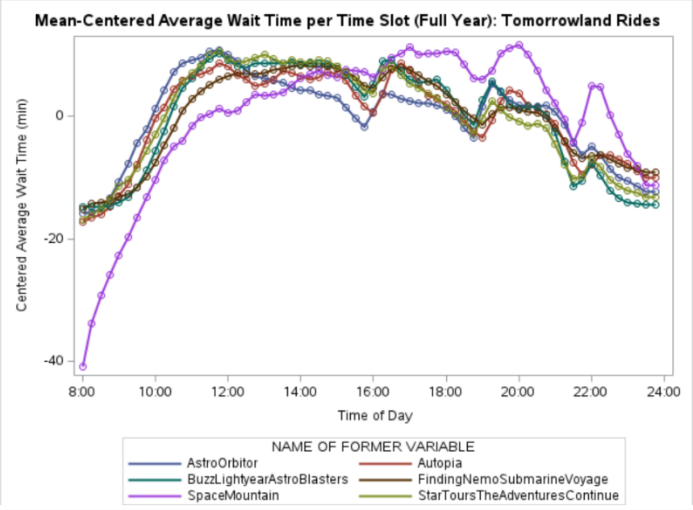


Figure 14. Mean-Centered Yearly Average Wait Time by Time of Day: Tomorrowland Rides

These views into how average wait times change within the park itself as a day progresses confirm a rational assumption: rides in the same area tend to have similar wait time distributions. The park overall has clear trends based on waves of traffic, but there is also local level change that differs based on region of the park. There is not a drastic difference between Tomorrowland and Fantasyland in particular, likely since they are not far from each other. However, we can see that there is a difference in the strength of the decrease in average wait times.

CONCLUSION

Looking at average wait time trends confirms much of what regular assumptions might say about planning a trip to Disneyland. However, having data confirm some of this common knowledge allows the creation of guidelines for what months fit best for an individual, as well as providing insight into how rides react to increasing park congestion. We find that, contrary to general assumptions, Summer has the lowest average ride wait times, although it is not much different to Spring. August is the best month of the year to plan a trip, particularly around the weekends, likely due to the end of grade school summer breaks. We determine four categorizations for rides based on their yearly average wait time distribution over and find that these categorizations match closely with a ride's intensity, theming, and whether the ride or its queue is exposed to open air. Finally, we analyze individual ride wait time distributions based on their regional proximity, as well as the rides with the highest average wait time. This confirms that areas of the park react similarly to crowds, and that rides of high perceived value can behave differently to the other rides in their areas, particularly if they are the more popular rides.

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